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Research Assistant, International Monetary Fund, Washington, DC, 2019 - 2021

Publications:

Boz et al. (2022). "Patterns of Invoicing Currency in Global Trade: New Evidence." *Journal of International Economics*, 136, 103604.

Working Papers:

"Workforce Demographics and New Software Adoption," with Sabrina Peng, Job Market Paper, November 2026.

Work In Progress:

"Perceived Education Returns Abroad and Investment at Home," with Emma Naslund-Hadley, Siu Yuat Wong, and Qianyao Ye, November 2026.

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Dissertation Abstract

Workforce Age Structure and Frontier Technology Adoption: Evidence from Newly Released Software, with Sabrina Peng [**Job Market Paper**]

Workforce aging is often viewed as an obstacle to economic growth because it reduces labor supply and raises old-age dependency. Aging may also slow growth through an understudied channel: the speed at which societies adopt new technologies. Older workers may avoid learning new technologies because they have shorter horizons over which to recoup the benefits, face higher opportunity costs from accumulated experience with previous tools, or incur higher direct learning costs. Thus, firms in older labor markets may be slower to adopt new technologies.

This paper estimates the causal effect of local workforce age composition on firms' adoption of newly released software technologies in the U.S. We use Lightcast job postings to track local employer demand for 15 software technologies released between 2012 and 2016. Each new technology is matched to a legacy counterpart, an older technology used for similar tasks, which allows for studying substitution from legacy to new tools within the same task family. Because firms need workers who can use these tools to adopt them, we interpret demand for frontier software skills as a revealed-demand measure of local technology adoption. Workforce age composition is measured by the local share of college-educated workers ages 50 to 69, reflecting the workforce most relevant for software-related skill demand. We instrument for the local old-worker share using a Bartik-style measure constructed from historical commuting-zone age composition and national survival rates.

Our estimates show that, within five years of release, employers in older labor markets post progressively fewer vacancies requiring the newest software technologies. A one-percentage-point increase in the older-worker share reduces the new technology's share of job postings by roughly 12 percent of its mean four years after release. Moreover, the age-related adoption gap is larger when the new software is more difficult to master. Difficulty is measured using an LLM-based technology-pair transition index whose scoring framework is informed by a focus group of software professionals. The index captures how difficult it is to move from legacy to new technologies and is validated against survey responses from computer scientists and software professionals.

To explain the empirical findings and evaluate policy counterfactuals, we develop and calibrate a parsimonious quantitative model linking workers' retraining decisions to firms' demand for frontier-technology skills. The model formalizes the three mechanisms that can make retraining less attractive for older workers: shorter remaining work horizons, greater accumulated experience with legacy technologies, and higher direct learning costs. We calibrate the model to match the age gradient in adoption estimated in our empirical analysis. The calibrated model shows that experience loss explains almost all of the age adoption gap, accounting for 88% of it, as opposed to 3% for cognitive decline. The model also shows that policy levers changing workforce age structure can have different implications for technology adoption. Reducing population growth to zero, as with extreme population aging, lowers aggregate adoption by about 5 percent. By contrast, a longer working horizon raises older workers' return to skill investment. Raising the retirement age by five years therefore almost fully offsets the drag that a larger share of older workers places on aggregate adoption.

Perceived Education Returns Abroad and Investment at Home, with Emma Naslund-Hadley, Siu Yuat Wong, and Qianyao Ye [**Work in Progress**]

Migration is a central economic strategy for many Central American households, including those whose children may later consider migrating to the United States. The possibility of migration can shape educational investment long before migration occurs by changing how households perceive

the returns to schooling. In a recent household survey conducted in El Salvador, we find that caretakers of children ages 8 to 12 substantially overestimate wages in the United States, especially for workers with secondary schooling, by more than 300 percent. Caretakers also overestimate the return to education in the United States relative to El Salvador, and these perceived U.S. returns are strongly associated with the amount of time parents spend helping children with schoolwork. These patterns suggest that beliefs about foreign labor-market opportunities shape human-capital investment at home.

Building on this evidence, the project studies how perceived returns to education abroad, perceived migration costs and barriers, and domestic education returns jointly affect children's schooling investments and household migration decisions. The empirical design uses a randomized information experiment with caretakers of school-age children in El Salvador. Caretakers receive experimentally varied information about U.S. wages for Salvadorans, migration success probabilities, wage uncertainty, and returns to education in El Salvador. The experiment measures how these information treatments change beliefs, migration intentions, and willingness to invest in supplemental education. The project tests a central mechanism in the brain-gain hypothesis: whether the possibility of migration raises education investment before migration occurs, and whether households respond more strongly to information about foreign earnings, migration risk, or domestic returns to schooling.